

Headline XIX: the phase-dressed leading term on the symmetric box

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Abstract

Unit 217 closes Astra #23's programme D2: interior coordinates with phase. On the orthant $x = \sigma \cdot u$ of $(-1, 1]^d$ one has $x^{2k} = u^{2k}$, $x^h = \chi_h(\sigma)u^h$ and $x^k \xi(x) = \varepsilon_\sigma u^k \xi(\sigma u)$ with $\chi_h(\sigma) = \prod_i \sigma_i^{h_i}$, $\varepsilon_\sigma = \prod_i \sigma_i^{k_i}$; by the generic reflection identity of unit 216 the symmetric-box dressed integral is the sum over orthants of unit-box chart integrals with phase $\varepsilon_\sigma \xi \circ \sigma$ and amplitude $\chi_h(\sigma) \eta \circ \sigma$, and Headline XIII on each orthant gives

$$\frac{\int_{(-1,1]^d} \eta x^h e^{-\beta N^2 x^{2k} + \beta N x^k \xi(x)} dx}{N^{-p}(\log N)^{m-1}} \rightarrow \sum_{\sigma \in \{\pm 1\}^d} \text{phaseCoeff}(\varepsilon_\sigma \xi \circ \sigma, \chi_h(\sigma) \eta \circ \sigma)$$

(headline_symmetric_phase_leading); for the absolute density $|x|^h$ the sign $\chi_h(\sigma)$ disappears (headline_symmetric_abs_phase_leading). Odd k_i flip the phase on the corresponding reflection ($J_p(a)$ against $J_p(-a)$): no parity cancellation is asserted in general, as Astra #22–23 warned; the zero-phase parity rule of unit 197 is the case $\xi = 0$. Zero sorry/axiom; 9094 jobs.

1 The things formalised

```
noncomputable def phaseSign (e : Fin d → ) ( : Fin d → Bool) : := i, sgn (
  i) ^ e i
theorem phaseSign_sq : phaseSign e ^ 2 = 1; theorem abs_sgn (b) : |sgn b| = 1
theorem prod_reflect_pow : i, reflect u i ^ e i = phaseSign e * i, u i ^ e i
theorem prod_abs_reflect_pow (hu : u unitBox d) : i, |reflect u i| ^ e i =
  i, u i ^ e i
theorem quadKernel_mul_sign (h : ^ 2 = 1) : quadKernel a ( * s) = quadKernel
  ( * a) s
theorem symBox_phase_eq_sum (h) (h) : x in symBox d, x * (( x^h) *
  quadKernel ( x) (N x^k))
  = , chartIntegral d h k N (fun u => phaseSign k * (reflect u)) (fun
    u => phaseSign h * (reflect u))
theorem symBox_abs_phase_eq_sum ... (amplitude fun u => (reflect u), density |x
  i| ^ h i)
theorem headline_symmetric_phase_leading (hk) (hl) (h) (hmin) (hatt) (h) (h) :
  Tendsto (fun N => ( x in symBox (m+1), x * (( x^h) * exp (-( N^2 x^(2k))
    + (N x^k) x)))
    / (N^(-(2l)) * log N ^ (multCount (ratioExp h k) l - 1))) atTop
  ( ( , phaseCoeff h k l (fun u => phaseSign k * (reflect u)) (fun u
    => phaseSign h * (reflect u))))
theorem headline_symmetric_abs_phase_leading ... ( ( , phaseCoeff h k l (fun u
  => phaseSign k * (reflect u)) (fun u => (reflect u))))
```

2 Mathematics

Purely algebraic on top of units 216 and 202: $(\sigma_i u_i)^{h_i} = \sigma_i^{h_i} u_i^{h_i}$, $|\sigma_i u_i|^{h_i} = u_i^{h_i}$ on the positive orthant, and $e^{-\beta(\varepsilon s)^2 + \beta a(\varepsilon s)} = e^{-\beta s^2 + \beta(\varepsilon a)s}$ for $\varepsilon^2 = 1$ move all signs into the phase and the amplitude. Each orthant is then a Headline XIII chart with continuous inputs (reflection is continuous), and finite sums of limits add.

3 Lean notes

- After `congr 1` the goal shows the per-orthant integrals unfolded (the limit theorem's function is the explicit integral, not `chartIntegral`); to rewrite with an identity stated via `chartIntegral`, `unfold chartIntegral at hs`; `beta_reduce at hs` first.
- `MeasurableSet.univ_pi fun _ => measurableSet_Ioc` discharges measurability of `symBox` without unfolding it.

4 Status

Programmes A, A2, B, B2, D, D2, G complete; Headlines VI–XIX. Remaining: review v15 (units 216–217), the hand-off report v3 with the explicit list of assumptions for application.